

S4

Sampling

Learning Objectives

In this chapter, we will

- (a) understand the concepts of population and simple random sample;
- (b) understand that the sample mean $\bar{X}$ is a random variable with $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}$;
- (c) use the fact that $\bar{X}$ has a normal distribution if X has a normal distribution;
- (d) apply the distribution of sample mean $N\left(\mu, \frac{\sigma^2}{n}\right)$ to solve statistical problems in real-world contexts;
- (e) apply the Central Limit Theorem on $\bar{X}$ when X does not follow a normal distribution when the sample size n is sufficiently large (e.g. $n \geq 30$).
- (f) understand the use of the unbiased estimates of the population mean and variance from a sample, including cases where data are given in summarised form $\sum x$ and $\sum x^2$, or $\sum(x-a)$ and $\sum(x-a)^2$.

4.1 [# Introduction](#)

Statistics is the science of data. Data are the numerical values containing some information. Statistical tools can be used on a data set to draw statistical inferences. These statistical inferences are in turn used for various purposes. For example, a government can use collected data for policy formulation for the welfare of the people, marketing companies can use the data from consumer surveys to improve the company and to provide better services to the customer, etc. Such data is obtained through sample surveys.

Why do we need sampling?

Consider a light bulb manufacturer who claims that his light bulbs can last at least 750 hours. It is not practical for the manufacturer to test every light bulb to find out if it can last this duration, as the test will either destroy or affect the performance of the light bulbs. A realistic option would be to collect a sample of light bulbs and test them.

Another reason for taking a sample is that not all elements in the population can be easily identified. For instance, in the study of television viewing behaviour during commercial breaks, it is difficult to identify all the television viewers for the study. Even if all the television viewers could be identified, it would be too expensive and too time consuming to interview every one of them.

The process of collecting a sample is known as **sampling**. Sampling is common in everyday life and if carried out correctly, it allows us to make fairly accurate inferences about a population without having to study every single element in that population.

4.2 [# Sampling Terminologies](#)

Let us define some terms that are commonly used in sampling.

Population

A population is a set of elements (individuals, families, organizations, groups or events) with the characteristic we wish to understand.

Random or Non-Random Sample

A sample is a subset of the population. Usually, the size of the sample is much less than the size of the population.

A sample for which every individual of the population must have an **equal chance of being selected** as a member of the sample and the selection of an individual is **independent** (i.e. has no effect on the chances) of the selection of any other individual is known as a random sample. Otherwise, it is a non-random sample.

For example, to find out the lifetime of a certain brand of battery, a set of 90 batteries are taken from the population. The set of all batteries manufactured can be taken to be the population. The 90 chosen batteries is a subset of the population. When each battery in the population has equal chance of being selected and each battery being selected is independent of another battery being selected, the set of 90 selected batteries is a random sample.

4.3 # Sampling Distributions

One of the primary goals for obtaining samples from a population is to allow a researcher to infer characteristics about a given population based on results observed from the samples.

Suppose we are able to take a random sample of 100 light bulbs and observe that the average lifetime of these light bulbs (sample mean) is 745 hours. We may thus conclude that 745 hours is a reasonable estimate of the average lifetime of all the light bulbs manufactured. But how reliable is it? How does this estimate vary if we change the sample to 1000 light bulbs for instance?

To answer these questions, we investigate the sampling distribution of the sample mean of these light bulbs. We will introduce two key terms for this section before defining what a sampling distribution is.

Population Parameter

It is a fixed value which represents a certain population characteristic.

The population mean is an example of a population parameter.

Population parameters are often denoted by Greek letters in lower case (e.g. population mean and population variance are denoted by μ and σ^2 respectively).

Sample Statistic

It is a quantity that is calculated from a sample taken from a population. Unlike a population parameter which is a fixed value, a statistic is a characteristic or measure of a sample and it is a random variable. The sample mean, the sample proportion and the sample variance are examples of sample statistics. Sample statistics are random variables and are commonly denoted by Roman letters in upper case (e.g. Sample mean is denoted by $\bar{X}$).

Sampling Distribution

A sampling distribution is a distribution that describes the variability in the values of a sample statistic for each of the possible samples from a population.

As mentioned earlier, the sample mean is an example of a sample statistic, and thus it has a (sampling) distribution. In the light bulb example, suppose we were to take a second random sample of size 100 and record its mean. It would not likely be exactly 745 hours but perhaps be close to that value, for example, 748 hours. If we repeat this process for a third sample, we might get a mean of 751 hours, as illustrated in Figure 1.

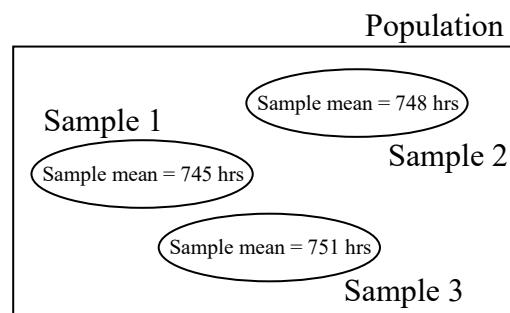


Figure 1

If we repeat this process many times and draw the histogram of these sample means of 100 light bulbs, the resulting histogram would indicate the distribution of the sample mean of the light bulbs.

There are other sample statistics and their associated distributions. However, in this chapter, we will focus only on the study of the distribution of the sample mean.

Example 1

Suppose that the random variable X takes values 1, 2 and 3 with equal probability. If a random sample of size 2 is selected with replacement, find the distribution of the sample mean $\bar{X}$.

Solution:

We denote a random sample of size 2 by (X_1, X_2) .

To find the probability distribution of $\bar{X}$ which is $\frac{X_1 + X_2}{2}$ in this case, we first consider all the possible outcomes for the average of X_1 and X_2 :

	x_1		
x_2	1	2	3
1	1	1.5	2
2	1.5	2	2.5
3	2	2.5	3

The probability distribution of $\bar{X}$ is tabulated as shown.

$\bar{x}$	1	1.5	2	2.5	3
$P(\bar{X} = \bar{x})$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{1}{3}$	$\frac{2}{9}$	$\frac{1}{9}$

□

Note

- In this case, the sample mean $\bar{X}$ is a discrete random variable and $\sum_{\text{all } \bar{x}} P(\bar{X} = \bar{x}) = 1$.
- Each value of $\bar{X}$, denoted as $\bar{x}$, is associated with a probability.

4.4 # Expectation and Variance of the Sample Mean

Since the sample mean $\bar{X}$ is a random variable, we can discuss about the expectation of the sample mean, $E(\bar{X})$ and its variance, $\text{Var}(\bar{X})$.

Important Result

Suppose $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population with mean μ and variance σ^2 , then the sample mean $\bar{X}$, where $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$, is such that

$$E(\bar{X}) = \mu \text{ and } \text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$$

Proof:

$$\begin{aligned} E(\bar{X}) &= E\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) = \frac{1}{n}(E(X_1) + E(X_2) + \dots + E(X_n)) \\ &= \frac{1}{n}(\mu + \mu + \dots + \mu) \\ &= \frac{1}{n}(n\mu) \\ &= \mu \end{aligned}$$

$$\begin{aligned} \text{Var}(\bar{X}) &= \text{Var}\left(\frac{X_1 + X_2 + \dots + X_n}{n}\right) \\ &= \frac{1}{n^2}(\text{Var}(X_1) + \text{Var}(X_2) + \dots + \text{Var}(X_n)) \quad (\text{since } X_i\text{'s are independent}) \\ &= \frac{1}{n^2}(\sigma^2 + \sigma^2 + \dots + \sigma^2) \\ &= \frac{1}{n^2}(n\sigma^2) \\ &= \frac{\sigma^2}{n} \end{aligned}$$

□

Note

1. The expected value, or the expectation, of the sample mean is equal to the population mean.
2. The variance of the sample mean is smaller than the population variance.

Example 2

Consider $\bar{X} = \frac{X_1 + X_2 + \dots + X_8}{8}$, where $X_1, X_2, \dots, X_8$ are 8 independent observations taken from each of the following distributions:

- (a) $B(7, 0.9)$
- (b) $N(30, 16)$

Calculate $E(\bar{X})$ and $\text{Var}(\bar{X})$ for each of the above cases.

Solution

(a) $E(\bar{X}) = \mu = np = 6.3$, $\text{Var}(\bar{X}) = \frac{\sigma^2}{8} = \frac{np(1-p)}{8} = 0.07875$ (EXACT Value no need to correct to 3 s.f.)

(b) $E(\bar{X}) = \mu = 30$, $\text{Var}(\bar{X}) = \frac{\sigma^2}{8} = \frac{16}{8} = 2$

□

4.5 # Distribution of the Sample Mean

In **Example 1**, we listed out all possible outcomes of $\bar{X}$ before finding its distribution. However, if the sample size is large or X takes many values, the method of listing out all possible outcomes and computing the probability of each outcome will be tedious. Fortunately, under certain conditions, the distribution of the sample mean follows a normal distribution or can be approximated to follow a normal distribution.

(A) Sample Obtained from a Normal Distribution

Recall from **3.7** that if $X_1, X_2, \dots, X_n$ are n independent normal random variables, then the sum $X_1 + X_2 + \dots + X_n$ is also normally distributed with $X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2)$.

Building from this result, if the population from which we draw our sample is normally distributed, then the distribution of the sample mean is also normally distributed:

Suppose $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a population which is normally distributed with mean μ and variance σ^2 , then the sample mean $\bar{X}$, where $\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$, also follows a normal distribution with $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$.

Example 3

The random variable X has a normal distribution with mean 60 and standard deviation 4. A random sample of 10 observations of X is taken. Find the probability that the sample mean is between 60 and 63.

80 samples, each of size 10, are now taken. How many of the samples would you expect the mean to lie between 60 and 63?

Solution

Given $X \sim N(60, 4^2)$. Then $\bar{X} \sim N\left(60, \frac{4^2}{10}\right)$.

Hence $P(60 < \bar{X} < 63) = 0.49115 = 0.491$

Let W be the number of samples that has sample mean between 60 and 63.

$$W \sim B(80, 0.49115)$$

Thus, $E(W) = 80 \times 0.49115 = 39.3$

□

Example 4

Random samples are taken from a normal distribution with mean 74 and variance 36. Determine the minimum size of the sample required if the probability that the sample mean exceeds 72 is at least 0.9.

Solution

Let X be normally distributed with mean 74 and variance 36.

Then $X \sim N(74, 36)$, and $\bar{X} \sim N\left(74, \frac{36}{n}\right)$.

$$P(\bar{X} > 72) \geq 0.9$$

Method 1: Standardising $\bar{X}$,

$$P\left(Z > \frac{72 - 74}{\sqrt{\frac{36}{n}}}\right) \geq 0.9 \Rightarrow \frac{-2}{\sqrt{\frac{36}{n}}} \leq -1.28155$$

$$\therefore -\frac{\sqrt{n}}{3} \leq -1.28155 \Rightarrow n \geq 14.78$$

Hence, the minimum size of the sample is 15.

Method 2: Using GC to find n which satisfies $P(\bar{X} > 72) \geq 0.9$:

- Press $\boxed{Y=}$ and in Y_1 , press $\boxed{2nd}\boxed{VARS}$ and select **2: normalcdf**.

Key in lower: 72, upper: $[EE]99$ $\mu: 74, \sigma: \frac{6}{\sqrt{X}}$.

- Press $\boxed{2nd}\boxed{TABLE}$ to go to TABLE.

- Scroll down the table till the first instance that the value of Y_1 exceeds 0.9 as shown. If n is a huge number, press $\boxed{2nd}\boxed{[TBLSET]}$ to change the starting value.

Starting value for n

Step size for n

TABLE SETUP
TblStart=9
 $\Delta Tbl=1$
Indpt: **Auto** Ask
Depnd: **Auto** Ask

NORMAL FLOAT AUTO re^(0i) DEGREE CL

Plot1	Plot2	Plot3
$\blacksquare Y_1 = \text{normalcdf}(72, E99, 74, 6/\sqrt{X})$		
$\blacksquare Y_2 =$		
$\blacksquare Y_3 =$		
$\blacksquare Y_4 =$		
$\blacksquare Y_5 =$		
$\blacksquare Y_6 =$		
$\blacksquare Y_7 =$		
$\blacksquare Y_8 =$		

NORMAL FLOAT AUTO re^(0i) RADIAN MP
PRESS + FOR ΔTbl

X	Y1			
9	.84134			
10	.85408			
11	.86554			
12	.87589			
13	.88529			
14	.89384			
15	.90165			
16	.90879			
17	.91534			
18	.92135			
19	.92688			

X=15

From GC,

when $n = 14$, $P(\bar{X} > 72) = 0.89384$

when $n = 15$ $P(\bar{X} > 72) = 0.90165$

Hence minimum size of the sample is 15.

(B) # Large Sample Obtained From Non-Normal or Unknown Distributions

Often, we do not know the distribution of the population from which our sample is taken. In such a situation, the sampling distribution would not be known to us too. Even in cases where we have knowledge of the distribution of the population, determining the sampling distribution may be an extremely difficult task to carry out.

The above scenarios posted great challenges for mathematicians in the 18th century. Through the works of many mathematicians, including de Moivre, Laplace and Poisson, a very important theorem known as the **Central Limit Theorem (CLT)** came into existence.

Subsequently, mathematicians such as Dirichlet, Bessel and Cauchy attempted to give a proof to this theorem but their proofs were somewhat incomplete. It was only in the late 19th century that the first complete proofs of this theorem were presented by Russian mathematicians such as Chebyshev, Markov and Liapounov.

Central Limit Theorem (CLT)

Suppose X is a random variable from a population of **non-normal or unknown distribution** with mean μ and variance σ^2 , and $X_1, X_2, \dots, X_n$ is a random sample of n independent observations of X . Then, for sufficiently large n ,

$$\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ approximately.}$$

Note

1. The larger the sample size n , the closer the distribution of the sample mean is to the normal distribution.
2. Sample size of at least 30 is considered sufficiently large.

Besides problems involving mean of samples, the Central Limit Theorem is also applicable to problems involving the sum of n independent observations of X .

Corollary to the Central Limit Theorem

Suppose X is a random variable from a population of **non-normal or unknown distribution** with mean μ and variance σ^2 , and $X_1, X_2, \dots, X_n$ is a random sample of n independent observations of X . Then, for sufficiently large n ,

$$X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2) \text{ approximately.}$$

Demonstration of Central Limit Theorem: (<https://tinyurl.com/CLTDemoApp>)

Example 5

- (a) It is given that the random variable X has a normal distribution with mean 30 and standard deviation 5.
- (i) The mean of a random sample of 10 values of X is denoted by $\bar{X}$. Find $P(\bar{X} > 31)$.
- (ii) Find the probability that the sum of a random sample of 10 values of X exceeds 305.
- (b) The random variable Y has a distribution with mean 30 and standard deviation 5.
- (i) The mean of a random sample of 100 values of Y is denoted by $\bar{Y}$. Find $P(\bar{Y} > 31)$.
- (ii) Find the probability that the sum of a random sample of 100 values of Y exceeds 3050.

Solution

- (a) (i) Given $X \sim N(30, 5^2)$, Then $\bar{X} \sim N\left(30, \frac{5^2}{10}\right)$.
Hence $P(\bar{X} > 31) = 0.264$.
- (ii) Let $X_1, X_2, \dots, X_{10}$ be the random sample of 10 values of X .
Then $X_1 + X_2 + \dots + X_{10} \sim N(300, 250)$
Hence $P(X_1 + X_2 + \dots + X_{10} > 305) = 0.376$
- (b) (i) Since $n = 100$ is large, by CLT, $\bar{Y} \sim N\left(30, \frac{5^2}{100}\right)$ approximately.
Hence required probability $= P(\bar{Y} > 31) = 0.0228$
- (ii) Let $Y_1, Y_2, \dots, Y_{100}$ be the random sample of 100 values of Y .
Since $n = 100$ is large, by CLT,
 $Y_1 + Y_2 + \dots + Y_{100} \sim N(3000, 2500)$ approximately.
Hence required probability $= P(Y_1 + Y_2 + \dots + Y_{100} > 3050) = 0.159$

Alternative Solution

Let $Y_1, Y_2, \dots, Y_{100}$ be the random sample of 100 values of Y .

$$E(Y) = 30, \text{Var}(Y) = 25.$$

Since $n = 100$ is large, by CLT, $\bar{Y} \sim N\left(30, \frac{5^2}{100}\right)$ approximately.

$$\text{Hence required probability} = P\left(\bar{Y} > \frac{3050}{100}\right) = 0.159$$

Example 6

A random sample of size 50 is taken from $B(10, 0.28)$. Find the probability that the sample mean exceeds 2.5.

Solution

Let $X \sim B(10, 0.28)$.

Then mean $\mu = np = 2.8$ and variance $\sigma^2 = np(1-p) = 2.016$

Since $n = 50$ is large, by CLT, $\bar{X} \sim N\left(2.8, \frac{2.016}{50}\right)$ approximately.

Hence required probability $= P(\bar{X} > 2.5) = 0.932$

Example 7

The content of a randomly chosen packet of lemon tea is found to have a mean of 200 ml and a standard deviation of 15 ml. The content of another randomly chosen packet of peach tea is found to have a mean of 150 ml and a standard deviation of 20 ml. If a random sample of 50 packets lemon tea and a random sample of 60 packets of peach tea are taken, what is the probability that the difference between the total amounts of content of the samples for each type is more than 800 ml?

Solution

Let $L_1, L_2, \dots, L_{50}$ be a random sample of 50 packets of lemon tea.

Let $P_1, P_2, \dots, P_{60}$ be a random sample of 60 packets of peach tea.

Given $E(L_i) = 200$, $\text{Var}(L_i) = 15^2$ for $i = 1, 2, \dots, 50$.

$E(P_i) = 150$, $\text{Var}(P_i) = 20^2$ for $i = 1, 2, \dots, 60$.

Let $S_L = L_1 + L_2 + \dots + L_{50}$ and $S_P = P_1 + P_2 + \dots + P_{60}$.

Since sample size for both samples are large (50 and 60 respectively), by CLT, $S_L \sim N(10000, 11250)$ approximately and $S_P \sim N(9000, 24000)$ approximately.

$\therefore S_L - S_P \sim N(1000, 35250)$ approximately.

Required probability $= P(|S_L - S_P| > 800) = P(S_L - S_P > 800) + P(S_L - S_P < -800) = 0.857$

Note:

$P(S_L - S_P > 800) \neq$

$P(S_L - S_P < -800)$

as the mean of $S_L - S_P$ is 1000, not zero.

4.6 # Estimation of Population Parameters

Statistical inference is the process of drawing conclusions about a population based on a random sample drawn from it.

In the previous section, we have learnt how the population mean and variance are used to find probabilities. However, in practice, the values of these population parameters are often unknown to us. Thus, it is essential to learn how to estimate population parameters from samples drawn from a population.

In this section, we will find out how the sample statistics, namely the sample mean and the sample variance, can be used to estimate the population mean and the population variance.

Suppose $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a large population.

Then

$$\text{Sample mean } \bar{X} = \frac{X_1 + X_2 + X_3 + \dots + X_n}{n} = \frac{\sum_{i=1}^n X_i}{n},$$

$$\text{Sample variance} = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 = \frac{1}{n} \left(\sum_{i=1}^n X_i^2 - \frac{\left(\sum_{i=1}^n X_i \right)^2}{n} \right).$$

Using $\sum X$ to denote $\sum_{i=1}^n X_i$, we have

$$\text{Sample mean } \bar{X} = \frac{\sum X}{n}$$

$$\text{Sample variance} = \frac{1}{n} \sum (X - \bar{X})^2 = \frac{1}{n} \left(\sum X^2 - \frac{(\sum X)^2}{n} \right)$$

4.7 # Point Estimation

Point Estimate

A (point) estimate of a population parameter is a single value used to estimate the parameter. Estimates are calculated values and they are denoted by lower case letter. For example, the sample mean $\bar{x}$ is a point estimate of the population mean μ .

Point Estimator

A (point) estimator of a population parameter is a statistic (formula) used to obtain the point estimate. Estimators are random variables and they are represented by upper case letters. For example, the sample mean $\bar{X}$ is an estimator of the population mean μ .

Instead of point estimation, we may use a range of values to estimate a population parameter of interest. This method is called interval estimation. In the subsequent sections, when we mention estimation or related terms, we will always refer to point estimation.

4.8 # Unbiased Estimation

Three marksmen are practising for a shooting competition and the results of one of their practice rounds are shown in Figure 2.

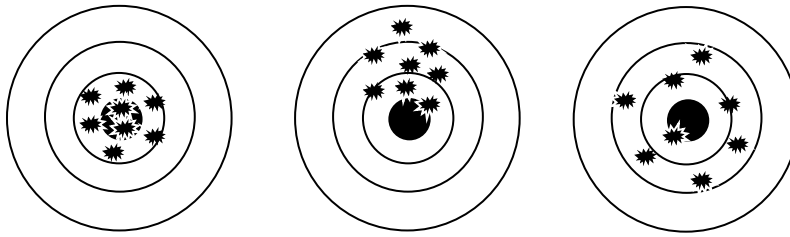


Figure 2

The target on the left has a tight cluster of shots which, on average, land on the bullseye. The target in the middle has all its shots clustered tightly together, but they are located above the bullseye. The target on the right has a large spread, but on average, the shots landed on the bullseye.

In statistical terms, we say that the shots on the target on the left are **unbiased** with a small variance, the shots on the target in the middle are **biased** with a small variance, and shots on the target on the right are unbiased with a large variance.

In general, the marksman shooting the target on the right will be deemed inferior in shooting than the other two marksmen because his shots are more varied.

Unbiased Estimator

A statistic is said to be an unbiased estimator of a population parameter if the mean of the sampling distribution of that statistic is equal to the value of the population parameter being estimated.

That is, if T is an estimator of a population parameter θ , and $E(T) = \theta$, then T is called an unbiased estimator of θ .

This definition is similar to finding the values of all the possible estimates computed from an estimator, and take the average. If this average value is equal to the population parameter, then we say that the estimator is an unbiased estimator of the population parameter.

Unbiased Estimate

The value obtained from the unbiased estimator of a population parameter is termed the unbiased estimate of the population parameter.

(A) Unbiased Estimator of the Population Mean

Let $X_1, X_2, \dots, X_n$ be a random sample taken from a population with unknown mean μ and $\bar{X}$ denotes the sample mean of this sample. Then, it follows from 4.4 that $E(\bar{X}) = \mu$ and thus $\bar{X}$ is an unbiased estimator of μ .

$$E(\bar{X}) = \mu$$

If we have obtained a specific sample and calculate its sample mean value $\bar{x}$, then $\bar{x}$ is called an unbiased estimate of the population mean. Note that $\bar{X}$ is a random variable as discussed in 4.5, whereas $\bar{x}$ is a single sample mean value of one particular sample.

(B) # Unbiased Estimator of the Population Variance

Let $X_1, X_2, \dots, X_n$ be a random sample taken from a population with unknown variance σ^2 .

Then the random variable $S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$ is an unbiased estimator of σ^2 . Refer to

Appendix A for a proof.

$$E(S^2) = \sigma^2$$

Similar to the case of the unbiased estimate of the sample mean, if we have obtained a specific sample and use the data values to calculate the quantity $\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$, then the value obtained, say s^2 , is called an unbiased estimate of the population variance.

Using $\sum x$ to denote $\sum_{i=1}^n x_i$, we have $s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2$. Expanding $\sum (x - \bar{x})^2$, we have

$$\begin{aligned} s^2 &= \frac{1}{n-1} \sum (x^2 - 2x\bar{x} + \bar{x}^2) \\ &= \frac{1}{n-1} \left(\sum x^2 - 2\sum x \left(\frac{\sum x}{n} \right) + n \left(\frac{\sum x}{n} \right)^2 \right) \\ &= \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) \end{aligned}$$

In MF 27, the unbiased estimate of the population variance is given by

$$s^2 = \frac{n}{n-1} \left(\frac{\sum (x - \bar{x})^2}{n} \right) = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Note

1. Comparing the formulae for sample variance (from 4.6) and that for s^2 (from 4.8), we note that the unbiased estimate of the population variance is related to the sample variance by the formula

$$s^2 = \frac{n}{n-1} (\text{sample variance})$$

2. Sample variance is NOT an unbiased estimator of the population variance σ^2 .
3. In the H2 mathematics syllabus, there is no specific notation for sample variance.

$$\text{Sample variance} = \frac{1}{n} \sum (x - \bar{x})^2 = \frac{1}{n} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$$

Example 8

- (a) A random sample of 20 observations $X_1, X_2, \dots, X_{20}$ was drawn from a large population and the following summarized data were obtained.

$$\sum x = 22.8, \quad \sum x^2 = 27.55.$$

Obtain the unbiased estimates of the population mean and variance.

- (b) Another random sample of 35 observations $X_1, X_2, \dots, X_{35}$ was drawn from the same large population and the sample mean and **sample variance** were computed to be 22.8 and 0.1 respectively. Find the unbiased estimates of the population mean and variance in this case.

Solution

- (a) Unbiased estimate of the population mean is $\bar{x} = \frac{\sum x}{n} = \frac{22.8}{20} = 1.14$

Unbiased estimate of the population variance is

$$s^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{1}{20-1} \left(27.55 - \frac{22.8^2}{20} \right) = 0.082$$

- (b) Given sample mean = 22.8 and sample variance = 0.1 .
Unbiased estimate of the population mean = sample mean = 22.8 .

Unbiased estimate of the population variance

$$s^2 = \frac{n}{n-1} (\text{sample variance}) = \frac{35}{34} (0.1) = 0.103$$

□

Example 9

A random sample of 10 Everbright bulbs was taken and their lifetimes, x hours, were obtained. The results were summarised by

$$\sum (x - 1000) = 1890 \quad \text{and} \quad \sum (x - 1000)^2 = 362050.2 .$$

Calculate the unbiased estimates of the mean and variance of the lifetimes of Everbright bulbs.

Solution

Let $y = x - 1000$

Thus $\sum y = 1890$ and $\sum y^2 = 362050.2$

$$\bar{y} = \frac{1890}{10} = 189$$

$$s^2 = \frac{1}{n-1} \left(\sum y^2 - \frac{(\sum y)^2}{n} \right) = \frac{1}{10-1} \left(362050.2 - \frac{1890^2}{10} \right) = 537.8$$

An unbiased estimate of the population mean = $189 + 1000 = 1189$ hours

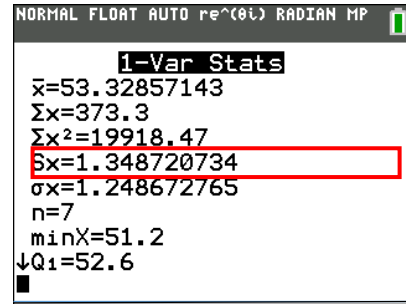
An unbiased estimate for the population variance = 538 hours²

□

$$\text{Hence } s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2 = \frac{1}{n-1} \sum (y - \bar{y})^2 = \frac{1}{n-1} \left(\sum y^2 - \frac{(\sum y)^2}{n} \right).$$

N-15

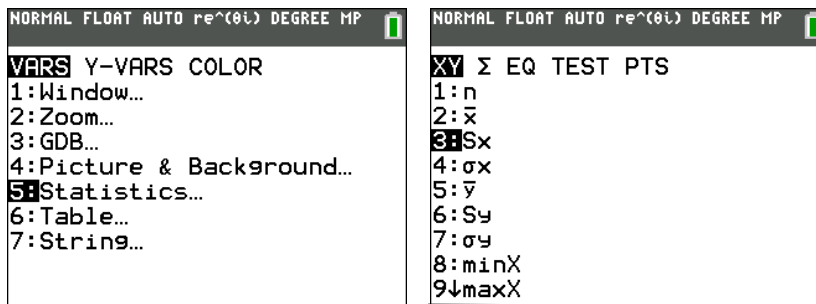
From the summarised results in the screenshot,
 sample variance $= (\sigma_x)^2 = (1.248672765)^2 = 1.56$,
 and the unbiased estimate of the population variance
 (which is what we want to find) is
 $(S_x)^2 = (1.348720734)^2 = 1.82$.
 Note that $(S_x)^2 > (\sigma_x)^2$.



Note

Instead of typing out the full string of numbers 1.348720734 for S_x , we can call out this value from our GC HOME SCREEN by doing the following steps.

- ① Press **[VARS]** and select **5: Statistics**.
- ② Select **3: S_x** and then press **[x²]** and **[ENTER]** to get the required value.



Summary

Notations

Population mean (fixed value)	Sample mean statistic (variable)	Sample mean value (calculated value)	Unbiased estimator of population mean (variable)	Unbiased estimate of population mean (calculated value)
μ	$\bar{X}$	$\bar{x}$	$\bar{X}$	$\bar{x}$

Population variance (fixed value)	Sample variance statistic (variable)	Sample variance value (calculated value)	Unbiased estimator of population variance (variable)	Unbiased estimate of population variance (calculated value)
σ^2	No specific notation	No specific notation	S^2	s^2

- Suppose X is a random variable from a population which is normally distributed with mean μ and variance σ^2 , and $X_1, X_2, \dots, X_n$ is a random sample of n independent observations of X . Then

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right), \text{ and } X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2).$$

- Suppose X is a random variable from a population of non-normal or unknown distribution with mean μ and variance σ^2 , and $X_1, X_2, \dots, X_n$ is a random sample of n independent observations of X .

Then, provided n is sufficiently large, by the Central Limit Theorem (CLT),

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n} \sim N\left(\mu, \frac{\sigma^2}{n}\right) \text{ approximately, and}$$

$$X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2) \text{ approximately.}$$

Suppose $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a large population. Then

$$\text{Sample mean } \bar{X} = \frac{\sum X}{n}$$

$$\text{Sample variance} = \frac{1}{n} \sum (X - \bar{X})^2 = \frac{1}{n} \left(\sum X^2 - \frac{(\sum X)^2}{n} \right)$$

- A (point) estimate of a population parameter is a value calculated from the sample data that we can consider being the most plausible value of the population parameter.
- A (point) estimator of a population parameter is a statistic or formula used to obtain a point estimate.
- A statistic T is called an unbiased estimator of the population parameter θ if $E(T) = \theta$. The value obtained from the unbiased estimator is called the unbiased estimate of θ .
- Sample mean $\bar{x}$ is an unbiased estimate of the population mean μ .
- s^2 is an unbiased estimate of the population variance σ^2 and it is given by

$$s^2 = \frac{1}{n-1} \sum (x - \bar{x})^2 = \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right) = \frac{n}{n-1} (\text{sample variance}).$$

Statistics Tutorial 4 – Sampling

- 1 Calculate unbiased estimates of the population mean and variance, respectively, for each of the following summarised data:

- (a) $n = 200$, $\sum y = 10000$ and $\sum y^2 = 524500$,
- (b) $n = 100$, $\sum (w - 100) = 52$ and $\sum (w - 100)^2 = 93$,
- (c) $n = 12$, $\sum x = 44.8$ and $\sum (x - \bar{x})^2 = 1.9467$,
- (d) $n = 60$, $\bar{x} = 35$ and $\sum (x - 35)^2 = 180$,
- (e) $n = 80$, $\sum x = 1064$ and Sample Variance = 46.

- 2 **RVHS/2022/C2/H2/BT2/Q9 modified.**

A school wishes to investigate its students' performance in a physical fitness test.

- (a)
 - (i) Name a suitable sampling method the school could use.
 - (ii) Suggest a reason why the method given in part (i) is chosen.
- (b) The school sends out a survey on its message board to collect responses.
 - (i) Comment on whether the responses collected are a population or sample.
 - (ii) Suggest a reason why this may not result in a random sample.
 - (iii) Suggest a reason why the school may choose to use this method anyway.
 - (iv) Describe a procedure that the school could use to select a random sample of the scores of 40 students.
- (c) The school obtains a random sample of 40 students. The score X is summarized as follows:

$$\sum x = 2247 \quad \sum x^2 = 129\,049$$

Calculate the unbiased estimates for the mean and variance of students' scores in the school. [3]

It is known that the national average for a student's score is 50 with a standard deviation of 8.5. A random sample of 30 students is taken. Find the probability that the mean of the sample lies within 0.5 of the population mean. [2]

- 3 **HCI 2018/BT2/11 modified**

Milk companies can have "Higher in Calcium" and/or "Lower in Fat" printed on the packaging of the milk they produce if their packets of milk meet the following criteria:

Substance	Quantity	Printed on package
Calcium	> 130 milligrams(mg) per 100 millilitres(ml) of milk	"Higher in Calcium"
Fat	< 1.5 grams(g) per 100 g of milk	"Lower in Fat"

- (a) Milk Company A states on its packaging that their milk contains 1.55 g of fat in 100 ml of milk. Given that 100 ml of milk weighs 103 g, can the company print “Lower in Fat” on the packaging of their milk? Justify your answer. [2]
- (b) A random sample of 40 packets of milk produced by Company A is taken, and the amount of calcium (in mg) per 100 ml of milk for each packet is summarised as follows:

Amount of calcium (in mg) per 100 ml	128	129	130	131	132	133	134	135	136	137	138
No. of packets	6	5	7	4	4	5	4	1	2	1	1

Find the sample mean amount of calcium per 100 ml of milk. Suggest why it is not advisable to allow Company A to have “Higher in Calcium” printed on the packaging of their milk based solely on the value of this sample mean. [2]

- (c) Find the unbiased estimate of the **population** variance. [1]

4 ASRJC/2020/JC2/CT/P2/Q2(i) modified

An advertising campaign is launched to promote the consumption of Oattie cereal amongst pre-schoolers. An advertising consultant states that the amount of sugar content in a box of 500g is at most 80g. In order to test this assertion, a random sample of 12 boxes were selected and their sugar content, in x grams, are summarised as follows:

$$\sum (x - 76) = 6.83, \quad \sum (x - 76)^2 = 15.132.$$

- (a) Explain in this context what is meant by a “random sample”. [1]
- (b) What assumption(s), if any, does the advertising consultant need to make about the population distribution of the sugar content in the cereal boxes? [1]
- (c) Suggest with a reason why, in this context, the given data is summarised in terms of $(x - 76)$ rather than x . [1]
- (d) Explain what is meant by ‘a sample statistic T is an unbiased estimator of a population parameter θ ’. [1]
- (e) Calculate the unbiased estimates of the population mean and population variance of the sugar content of the cereal boxes. [2]

5 TJC JC2 Prelim 9758/2019/02/Q10a modified

Two random samples of different sample sizes of households in the town of Aimek were taken to find out the mean number of computers per household there. The first sample of 50 households gave the following results.

$$\sum x = 98, \quad \sum x^2 = 254,$$

where x is the number of computers in a household.

The results of the second sample of 60 households were summarised as follows.

$$\sum y = 118, \quad \sum y^2 = 314,$$

where y is the number of computers in a household.

- (i) By combining the two samples, find unbiased estimates of the population mean and variance of the number of computers per household in the town. [4]
- (ii) Describe what you understand by 'population' in the context of this question. [1]

6 TJC/2021/C2 BT2/P2/8 modified

There are 25 JC1 classes of different class sizes in a junior college. The heights of the students are recorded. The mean height of the JC1 students is 165 cm and the standard deviation is 6.3 cm.

- (a) Mr Lim is the teacher-in-charge of an inter-college friendly match. He needs to select 50 JC1 students for the match. He decides to randomly select 2 students from each of the 25 classes.
 - (i) Jane is a student in a class of n students. Find, in terms of n , the probability that Jane is one of the 2 students randomly selected from her class. [1]
 - (ii) Determine with a reason if the sample selected using Mr Lim's method is a random sample. [1]
- (b) A random sample of 50 JC1 students is selected. Find the probability that the mean height of the students in the sample is greater than 166 cm. [3]
- (c) It is given that there is a probability of more than 0.15 of drawing a random sample of n students whose mean height is greater than 166 cm. Find the **least** value of n . State any assumption, where applicable. **largest** [3]
- (d) Another random sample of 50 JC1 students is selected. Find the probability that the difference in mean height of the 2 samples is at least 2 cm. [3]

7 SAJC Prelim 9758/2021/02/Q11 modified

- (a) Anne, a Bubble Tea (BBT) seller, intends to increase the sales of BBT using a drink vending machine which delivers BBT into a cup when cash payment is made into the machine. The volume of BBT dispensed is normally distributed with mean 210 ml and standard deviation 5 ml. The capacity of a cup is 220 ml and the nominal amount of BBT in a cup is stated as 212 ml.
 - (i) Anne received complaints from some customers that there is a high proportion of cups with less than the nominal amount of BBT. It is assumed that the standard deviation of the volume of BBT dispensed is fixed while the mean volume of BBT dispensed could be adjusted. Find the range of the mean volume of BBT dispensed such that not more than 10% of the cups will contain less than the nominal amount of BBT. [3]

- (ii) Another customer bought n cups of BBT. Find the approximate probability that the total volume of BBT dispensed exceeds $211n$ ml as n becomes very large. [2]
- (iii) Anne wishes to gather feedback about her BBT. She decided to interview 50 customers who bought BBT from the vending machine during lunch time.
Give a reason as to whether Anne would obtain a random sample of customers. [1]
- (b) On another occasion, Anne deployed a staff to operate a BBT counter at a carnival. The average volume of BBT per serving is 200 ml and the standard deviation of the volume of BBT is given to be 10 ml.
- (i) Find the probability that the mean volume of 60 servings of BBT prepared by the staff is less than 198 ml. [3]
- (ii) Anne decides to obtain six samples, each of 60 servings of BBT prepared by the staff. Calculate the probability that exactly one out of the six samples will have their respective mean volume being less than 198 ml. State two assumptions made in this calculation. [3]

8 TJC Prelim 9758/2018/02/Q8

Box A contains five cards numbered 1, 1, 2, 2 and 3 and Box B contains three cards numbered 4, 4 and 5. Cards numbered 1 and 5 are red, whereas cards numbered 2, 3 and 4 are blue.

A card is drawn from each of the two boxes. If both cards are the same colour then the score will be the product of the numbers on the two cards. If both cards are different in colour then the score will be the sum of the numbers on the two cards.

Let X be the score obtained. The probability distribution of X is given below:

x	5	7	8	12
$P(X=x)$	h	k	$\frac{1}{3}$	k

- (i) Show that $h = \frac{2}{5}$. [2]
- (ii) Find k , $E(X)$ and $\text{Var}(X)$. [4]
- (iii) Find the probability that the mean score of 60 independent observations of X lies between 6.5 and 7.5. [3]

In a game, a player pays \$20 and draws a card each from Box A and Box B . She receives, in dollars, an amount equal to three times the score obtained. Find the variance of her winnings per game. [2]

9 RVHS Prelim 9758/2021/02/Q9

During each round of practice, John, does 10 multiple-choice questions and his score X , is the number of questions he answered correctly. On average, he has an 85% chance of answering each multiple-choice question correctly.

- (i) State in the context of the question, one assumption needed to model X by a binomial distribution. [1]

On a particular day, John does 4 rounds of practice.

- (ii) Find the probability that the total score for John in the 4 rounds of practice is more than 36. [2]
- (iii) Find the probability that John obtains a score of at most 9 in each of his first 2 rounds of practice. [3]

To qualify for an award, John will need a mean score of at least 8.8 for all his 50 rounds of practice.

- (iv) Find the probability that John will qualify for the award. [3]

10 EJC 2019/BT2/II/8 modified

A private-hire vehicle firm, Euber, employs drivers to operate a fleet of cars. The mileage covered by an Euber driver on a weekday is denoted by M km. A random sample of 60 Euber drivers is taken and their mileage can be summarised by

$$\sum (m - 250) = 540 \text{ and } \sum (m - 259)^2 = 8176.$$

- (i) Calculate unbiased estimates of the population mean and variance. [2]

~~Assume that the population mean and variance follow the answers calculated in part (i).~~

Random samples of n Euber drivers are taken, where n is large.

- ~~(ii) Over a long period of time, it is found that 25% of the samples have a mean mileage of at least 260 km. Find the value of n . [3]~~
- (iii) Explain whether it is necessary to assume that M follows a normal distribution in order for your calculations in part (ii) to hold. [1]
- (iv) It is now given that M is normally distributed with population variance 5^2 . Estimate the sample size needed so that Euber can be 95% certain that the sample mean mileage will differ from the true mean by less than 500m. [2]

Numerical Answers:

- 1 (a) 50, 123 (b) 100.52, 0.666 (c) 13.3, 46.6
(d) 3.73, 0.177 (e) 35, 3.05
- 2 (c) 56.175, 72.4; 0.253 3 (b) 131.45 (c) 7.28
- 4 (e) 76.6, 1.02 5 (i) $\frac{108}{55}$, $\frac{7912}{5995}$
- 6 (ai) $\frac{2}{n}$ (b) 0.131 (c) 42 (d) 0.112
- 7 (i) $\frac{218}{219} \leq \mu \leq 220$ (ii) 0 (bi) 0.0607 (ii) 0.266
- 8 (ii) $k = \frac{2}{15}$; $E(X) = \frac{36}{5}$; $\text{Var}(X) = \frac{392}{75}$ (iii) 0.836; 47.04
- 9 (ii) 0.130 (iii) 0.645 (iv) 0.0301
- 10 (i) $\frac{159}{259}$, 139 (ii) 63 (iv) 384

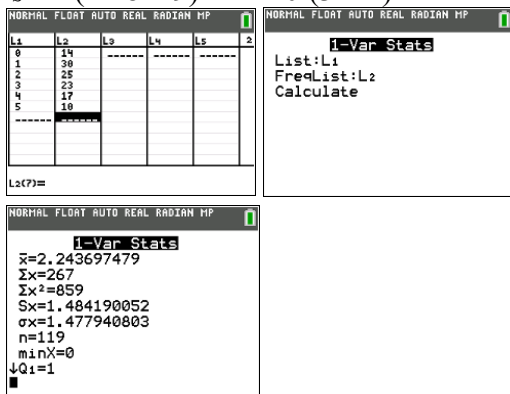
Skill Set

Chapter S4 Sampling

No.	Skills	Examples of questions involving the skills
1	Understand the concepts of population (a) population (b) sample, (c) random sample and (d) non-random sample.	E.g. In a certain county, there are 100 professional football clubs, arranged in 4 divisions. There are 22 clubs in Division One, 24 in Division Two, 26 in Division Three and 28 in Division Four. (i) Alice wishes to find out about approaches to training by clubs in Division One, so she sends a questionnaire to the 22 clubs in Division One. Explain whether these 22 clubs form a sample or a population. [1] (ii) Dilip wishes to investigate the facilities for supporters at the football clubs, but does not want to obtain the detailed information necessary from all 100 clubs. Explain how he should carry out his investigation, and why he should do the investigation in this way. [2] Solution (i) Since Alice wants to find out only about clubs in Division One, the 22 clubs in Division One will form the population. (ii) He should randomly pick a subset of clubs out of the 100 clubs in the 4 Divisions to obtain his information. In this way, every club has a equal chance of being selected, and so his sample is selected independently and it is not biased.
2	(a) If X has a normal distribution then $\bar{X}$ has a normal distribution i.e. If $X \sim N(\mu, \sigma^2)$, then $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right).$ (b) When X may not follow a normal distribution, apply the	E.g. (a) It is given that the random variable X has a normal distribution with mean 30 and standard deviation 5. (i) The mean of a random sample of 10 values of X is denoted by $\bar{X}$. Find $P(\bar{X} > 31)$. (ii) Find the probability that the sum of a random sample of 10 values of X exceeds 305. (b) The random variable Y has a distribution with

No.	Skills	Examples of questions involving the skills
	Central Limit Theorem on $\bar{X}$ i.e. If $X \sim$ unknown distribution with population mean μ and population variance σ^2, and $X_1, X_2, \dots, X_n$ is a random sample of n independent observations of X, then for sufficiently large n, $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately OR we can also write $X_1 + X_2 + \dots + X_n \sim N(n\mu, n\sigma^2)$ approximately.	mean 30 and standard deviation 5. (i) The mean of a random sample of 100 values of Y is denoted by $\bar{Y}$. Find $P(\bar{Y} > 31)$. (ii) Find the probability that the sum of a random sample of 100 values of Y exceeds 3050. Solution (a) (i) Given $X \sim N(30, 5^2)$, then $\bar{X} \sim N\left(30, \frac{5^2}{10}\right)$. $P(\bar{X} > 31) = 0.264$. (ii) Let $X_1, X_2, \dots, X_{10}$ be the random sample of 10 values of X. Then $X_1 + X_2 + \dots + X_{10} \sim N(300, 250)$ $P(X_1 + X_2 + \dots + X_{10} > 305) = 0.376$ (b) (i) Since $n = 100$ is large, by Central Limit Theorem, $\bar{Y} \sim N\left(30, \frac{5^2}{100}\right)$ approximately. $P(\bar{Y} > 31) = 0.0228$ (ii) Let $Y_1, Y_2, \dots, Y_{100}$ be the random sample of 100 values of Y. Since $n = 100$ is large, by Central Limit Theorem, $Y_1 + Y_2 + \dots + Y_{100} \sim N(3000, 2500)$ approximately. $P(Y_1 + Y_2 + \dots + Y_{100} > 3050) = 0.159$
3	A statistic is said to be an unbiased estimator of a population parameter if the mean of the sampling distribution of that statistic is equal to the value of the population parameter being estimated i.e. If T is an estimator of a population parameter θ, and $E(T) = \theta$, then T is an unbiased estimator of θ. (a) Unbiased Estimator for the population mean The random variable $\bar{X}$ is an	E.g. Calculate unbiased estimates of the population mean and variance, respectively, for each of the following summarised data: (a) $n = 200$, $\sum y = 10000$, $\sum y^2 = 524500$; (b) $n = 100$, $\sum (w - 100) = 52$, $\sum (w - 100)^2 = 93$; (c) $n = 80$, $\sum x = 1064$, Sample Variance = 46; (d) $n = 12$, $\sum x = 44.8$, $\sum (x - \bar{x})^2 = 1.9467$;

No.	Skills	Examples of questions involving the skills
	unbiased estimator of the population mean μ. If we have obtained a specific sample and calculate its sample mean value $\bar{x}$, then $\bar{x}$ is called an unbiased estimate of the population mean. (b) Unbiased Estimator for the population variance. The random variable S^2 is an unbiased estimator of the population variance, σ^2. If we have obtained a specific sample and calculate $s^2 = \frac{n}{n-1}(\text{sample variance}), \text{ then}$ s^2 is called an unbiased estimate of the population variance. In MF27, $s^2 = \frac{n}{n-1} \left(\frac{\sum (x - \bar{x})^2}{n} \right)$ $= \frac{1}{n-1} \left(\sum x^2 - \frac{(\sum x)^2}{n} \right)$ Calculate the unbiased estimates of the population mean and variance, including cases where the data are given in - summarised forms $\sum x$ and $\sum x^2$ or sample variance or $\sum (x - \bar{x})^2$ $\sum (x - a)$ and $\sum (x - a)^2$ - frequency table	Solution (a) $\bar{y} = \frac{10000}{200} = 50.$ $s^2 = \frac{1}{n-1} \left(\sum y^2 - \frac{(\sum y)^2}{n} \right)$ $= \frac{1}{200-1} \left(524500 - \frac{10000^2}{200} \right) = \frac{24500}{199}$ (b) $\bar{w} = \frac{52}{100} + 100 = 100.52$ $s^2 = \frac{1}{n-1} \left\{ \sum (w-100)^2 - \frac{[\sum (w-100)]^2}{n} \right\}$ $= \frac{1}{100-1} \left(93 - \frac{52^2}{100} \right) = \frac{1649}{2475}$ (c) $\bar{x} = \frac{1064}{80} = \frac{133}{10} = 13.3.$ $s^2 = \frac{n}{n-1} (\text{Sample Variance})$ $= \frac{80}{80-1} (46) = \frac{3680}{79}$ (d) $\bar{x} = \frac{44.8}{12} = \frac{56}{15} = 3.73$ $s^2 = \frac{\sum (x - \bar{x})^2}{11} = \frac{1.9467}{11} = 0.177$

No.	Skills	Examples of questions involving the skills														
		E.g. A coin is chosen at random from a population of recently produced coins. Let X be the age, in years, of the coin. A random sample of 119 observations of X was taken and the results can be summarised as follows: <table><tr><td>Age (x)</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td></tr><tr><td>Frequency (f)</td><td>14</td><td>30</td><td>25</td><td>23</td><td>17</td><td>10</td></tr></table> (i) Explain what is meant by ‘a sample statistic T is an unbiased estimator of a population parameter θ’. [1] (ii) Find the unbiased estimate of the population mean and variance. [2] Solution (i) A sample statistic T is an unbiased estimator of a population parameter θ if $E(T) = \theta$. (ii) $\bar{x} = 2.24$ (3 s.f.) $s^2 = (1.48419)^2 = 2.20$ (3 s.f.) 	Age (x)	0	1	2	3	4	5	Frequency (f)	14	30	25	23	17	10
Age (x)	0	1	2	3	4	5										
Frequency (f)	14	30	25	23	17	10										
Other calculations related to $\bar{X}$ may require application of skills acquired in S1 – S3.																

Supplementary Exercise

- 1 The heights of a new variety of sunflower can be modelled by a normal distribution with mean 2 m and standard deviation of 40 cm.
- (i) A random sample containing 50 sunflowers is taken and the mean height calculated. What is the probability that the sample mean lies between 195 cm and 205 cm?
 - (ii) A hundred such samples, each of 50 observations, are taken. In how many of these would you expect the sample mean to lie between 195 cm and 205 cm?

2 NYJC23/Prelim/II/Q9

The manufacturer of a particular toy drone claims that its average flying time (AFT) is 12 minutes. The quality control manager suspects that the AFT is different from what the manufacturer has claimed and decides to carry out a hypothesis test on a sample of these drones.

The flight durations, x minutes, of a random sample of 100 drones are summarised as follows.

$$\sum x = 1171 \quad \sum x^2 = 14023$$

- (a) State what it means for a sample to be random in this context. [1]
- (b) Calculate unbiased estimates of the population mean and variance of the AFT of the toy drones. [2]

3 SAJC17/C2BT2/II/Q6 (modified)

A coin is chosen at random from a population of recently produced coins. Let X be the age, in years, of the coin. A random sample of 119 observations of X was taken and the results can be summarised as follows:

Age (x)	0	1	2	3	4	5
Frequency (f)	14	30	25	23	17	10

- (i) Explain what is meant in this context by the term ‘a random sample’. [1]
- (ii) Calculate the mean age of the **sample** and its standard deviation, correct to 5 significant figures. [2]
- (iii) Explain what is meant by ‘a sample statistic T is an unbiased estimator of a population parameter θ ’. [1]
- (iv) Find the unbiased estimate of the **population** mean and variance. [2]

4 RVHS22/Prelim/II/Q12

The mass of a “Comforta” cushion has a normal distribution with mean 170g and standard deviation 12g.

- (i) Find the probability that the mass of a randomly chosen “Comforta” cushion is more than 165g. [1]
- (ii) Find the probability that, of three randomly chosen “Comforta” cushions, the mass of the heaviest cushion is less than 165g. [2]

Each “Comforta” cushion comes with a cushion cover. The mass of a cushion cover has a normal distribution with mean 20g and standard deviation σ .

(iii) It is given that more than 90% of “Comforta” cushions with their covers have mass at least 170g. Find the range of values of σ . [4]

(iv) Given that $\sigma = 8$. A “Comforta” cushion costs \$0.05 per gram to produce while a cushion cover costs \$0.02 per gram to produce. A “Comforta” cushion with cover is sold at \$10 each. Find the probability that profit will be made with the sale of a “Comforta” cushion with cover. [3]

The company manufacturing “Comforta” cushions decided to launch a new product called “Serena” cushions. The mean mass of a “Serena” cushion is 250g. A sample of 50 “Serena” cushions were collected for quality check and it was found that the sample standard deviation is 17g. Find the probability that the sample mean mass of a “Serena” cushion is at most 245g. [3]

5 ACJC17/BT2/I/Q10 (modified)

In a duck farm, a duckling infected with a particular virus has a 92% chance of survival.

(i) In a random sample of 30 ducklings infected with the virus, find the probability that at least two of them will not survive. [2]

(ii) In 80 such samples of 30 ducklings, find the probability that the mean number of ducklings per sample of 30 ducklings that do not survive is more than 2.5. [3]

6 RI Prelim 9758/2020/02/Q11(Modified)

In conjunction with the Great Singapore Sale, a certain electronics store is having a lucky draw for their customers. In each round of the lucky draw, the customer draws two balls randomly, one after another, with replacement from a box containing 20 red balls, 30 blue balls and 50 white balls. The colour of each ball drawn is noted and points are awarded accordingly as follows.

Colour of ball	Point(s)
Red	5
Blue	4
White	1

The customer’s score in each round of the lucky draw is the total number of points awarded for the balls drawn. To illustrate, a customer scores a total of 5 points if he draws a blue ball and a white ball in a round of the lucky draw, regardless of the order of appearance of the balls.

You may assume that the 100 balls are indistinguishable from each other apart from their colour.

Let X denote the total number of points scored by a customer in a round of the lucky draw.

(i) Tabulate the probability distribution of X . [3]

(ii) Find $E(X)$ and show that $\text{Var}(X) = 6.02$. [2]

Mr Lim participated in 50 rounds of the lucky draw.

Using a suitable approximation, find the probability that Mr Lim's average score is at least 6. [3]

- 7 A circular card, with a pointer pivoted at the centre is divided into 5 unequal sectors numbered "1", "2", "3", "4" and "5". The pointer is spun and the score will be the number at which the pointer stopped at. The probability of scoring a "5" is $1 - q$. The pointer is spun 10 times independently and the number of "5"s obtained is denoted by Y . Given that $\text{Var}(Y) = [E(Y)]^2$, show that $q = \frac{10}{11}$.

Supposed that 50 people are invited to spin the pointer 10 times each, find the probability that the mean number of times they obtain a "5" exceeds 1.

8 **TJC19/C2BT2/II/Q6**

The weight X kg of sand in a bag can be modelled by a distribution with mean 1.2 kg and standard deviation 0.4 kg. A random sample of n bags is taken where n is assumed to be large.

- (i) Explain why the mean weight of sand $\bar{X}$ in a random sample n bags is approximated by a normal distribution, and state the approximate distribution of $\bar{X}$. [2]
- (ii) Another random sample of $2n$ bags is taken. Determine which sample has a greater probability of having a mean weight of sand within 0.1 kg of 1.2 kg. [2]
- (iii) Ten random samples of 100 bags each are taken. Find the probability that **each** sample has a total weight of more than 121 kg of sand. [4]

9 **TJC Prelim 9758/2023/02/Q8**

The random variable X has the probability function

$$P(X = x) = \begin{cases} kx & \text{for } x = 3, 4, 5, \\ k(11 - x) & \text{for } x = 6, 7, 8. \end{cases}$$

where k is a constant.

(a) Show that $k = \frac{1}{24}$. [1]

(b) Find the exact values of $E(X)$ and $\text{Var}(X)$. [3]

(c) Fifty independent observations of X are taken. Using a suitable approximation, estimate the probability that the mean of these observations is at less than 4.9. [2]

10 **MJC14/Prelim/II/Q10**

The masses, in kilograms, for a randomly chosen pumpkin and a randomly chosen watermelon, are normally distributed with means 4.2 and 8.5, and standard deviations 1.6 and 2.2 respectively.

(i) Find the range of values of k such that the average mass of five randomly chosen watermelons exceeds k kilograms occurs at most 10% of the time. [3]

(ii) Find the probability that the total mass of five randomly chosen pumpkins differs from three times the average mass of five randomly chosen watermelons by more than 5 kg. [4]

(iii) A fruit store has 70 batches, each consisting of n pumpkins. It is known that, on average, 3 in 20 pumpkins are rotten. Given that in the 70 batches, the probability of having an average of at least 4 rotten pumpkins per batch is more than 0.7. Determine the minimum value of n . [4]

11 The 'reading age' of children about to start a secondary school is a measure of how good they are at reading and understanding printed text. A child's reading age, measured in years, is denoted by the random variable X . The distribution of X is assumed to be $N(\mu, \sigma^2)$.

(a) The reading age of a random sample of 20 children were measured and the data obtained is summarised by

$$\sum x = 232.6, \quad \sum x^2 = 2756.22.$$

Calculate unbiased estimates of μ and σ^2 , giving your answers correct to 2 decimal places.

(b) In order to obtain a more accurate estimate of μ , it is proposed to take a larger sample. Estimate the sample size needed so that we can be 95% certain that the sample mean reading age will differ from the true mean reading age by less than

6 months. Assume that it is known from previous studies that $\sigma^2 = 2.25$.

12 YIJC Prelim 9758/2021/02/Q10

A factory produces a large number of bottles. Based on past records, 4% of the bottles are defective. A departmental store manager wishes to purchase bottles from the factory. To decide whether to accept or reject a batch of bottles, the manager designs a sampling process. He takes a random sample of 30 bottles. The batch is accepted if there are no defective bottles and is rejected if there are more than 2 defective bottles. Otherwise, a second random sample of 30 bottles is taken. The batch is then accepted if there are fewer defective bottles in the second sample and is rejected otherwise.

- (i) Find the probability of accepting a batch of bottles. [2]
- (ii) If a batch is accepted, find the probability that there are at most 2 defective bottles found in the sampling process. [3]

Another departmental store manager purchases n randomly chosen boxes of 30 bottles each.

- (iii) Find the probability that a box has exactly 20 fewer defective bottles than non-defective bottles. [2]
- (iv) Find the greatest value of n if the probability that there are at least 2 boxes with exactly 20 fewer defective bottles than non-defective bottles is at most 0.05. [3]
- (v) Given that $n = 50$, find the probability that the mean number of defective bottles in a box is less than 1. [3]

13 RI14/Prelim/II/Q12 (modified)

The masses, in grams (g), of chicken and duck eggs are modelled as having independent normal distributions with means and standard deviations as shown in the table below.

	Mean (g)	Standard deviation (g)
Chicken Eggs	50	2
Duck Eggs	65	3

Chicken eggs are packaged in cartons of 6 and duck eggs are packaged in cartons of 4. It is given that the mass of the cartons are negligible.

- (i) Find the probability that the total mass of two randomly chosen cartons of chicken eggs exceeds twice the mass of a randomly chosen carton of duck eggs by at least 100 g. [3]

A random sample of n cartons of duck eggs are chosen and their mean mass M in grams(g) per carton is recorded.

- (ii) Given that $n = 4$, find $P(M > 265)$. [2]
- (iii) Find the approximate value of $P(M > 265)$ as n becomes large. [2]

- 14 PJC17/Prelim/II/Q8**
 Statistics has shown that 29% of the adult population (above 25 years old) in Singapore attained a university degree. A random sample of 50 such Singaporean adults is obtained.
- (i) Find the probability that more than 18 adults attained a university degree. [2]
 - (ii) Find the greatest value of m such that the probability that there are at least m adults who attained a university degree exceeds 0.9. [3]
- In a large scale statistical study, 80 such samples of 50 adults are taken. Find the probability that the average number of adults in each sample who attained a university degree does not exceed 14. [3]
- 15 NJC15/Prelim/II/Q9**
 In this question, you should state clearly the values of the parameters of any normal distributions you use.
 The weight, in grams, of a handphone is a random variable with the distribution $N(\mu, 5^2)$, where μ is a constant. The weight, in grams, of a tablet is a random variable with the independent distribution $N(330, 6^2)$.
- (a) Suppose $\mu = 130$. Calculate the probability that the total weight of 5 randomly selected handphones is less than twice the weight of one randomly selected tablet. [3]
 - (b) Suppose instead that μ is unknown. A random sample of n handphones is taken. There is a probability of at most 0.04 that the mean weight of these handphones differs from the population mean weight of the handphones by more than 1 gram. Calculate the least value of n . [4]

Numerical Answers:

- 1 (i) 0.623 (ii) 62.3
- 2 (b) 11.71, 3.14
- 3 (ii) $\bar{x} = 2.2437$ and Standard deviation of sample = 1.4779
- 4 (i) 0.662 (ii) 0.0388 (iii) $\sigma < 9.98$ (iv) 0.962, 0.0198
- 5 (i) 0.704 (ii) 0.274
- 6 (ii) 5.4; $P(\bar{X} \geq 6) = 0.0419$
- 7 0.240
- 8 (ii) $2n$ (iii) 0.000108
- 9 (b) $\frac{29}{12}$ (c) 0.00317
- 10 (i) $k \geq 9.76$ (ii) 0.477 (iii) 28
- 11 (a) 11.63; 2.69 (b) 35
- 12 (i) 0.549 (ii) 0.851 (iii) 0.00526 (iv) 67 (v) 0.0938
- 13 (i) 0.0745 ; (ii) 0.0478 ;
- 14 (i) 0.108 (ii) 10; 0.0817
- 15 (a) 0.729 (b) 106

Appendix A

To prove $E(S^2) = \sigma^2$,

$$\begin{aligned} E(S^2) &= E\left(\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2\right) \\ &= \frac{1}{n-1} E\left(\sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2)\right) \\ &= \frac{1}{n-1} E\left(\sum_{i=1}^n (X_i^2 - 2X_i\bar{X} + \bar{X}^2)\right) \\ &= \frac{1}{n-1} E\left(\sum_{i=1}^n X_i^2 - 2\bar{X} \sum_{i=1}^n X_i + \sum_{i=1}^n \bar{X}^2\right) \\ &= \frac{1}{n-1} E\left(\sum_{i=1}^n X_i^2 - 2\bar{X}(n\bar{X}) + n\bar{X}^2\right) \\ &= \frac{1}{n-1} E\left(\sum_{i=1}^n X_i^2 - n\bar{X}^2\right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n E(X_i^2) - nE(\bar{X}^2)\right) \\ &= \frac{1}{n-1} \left(\sum_{i=1}^n (\sigma^2 + \mu^2) - n\left(\frac{\sigma^2}{n} + \mu^2\right)\right) \\ &= \frac{1}{n-1} (n(\sigma^2 + \mu^2) - \sigma^2 - n\mu^2) \\ &= \sigma^2 \end{aligned}$$

Recall:

$$\text{Var}(X) = E(X^2) - \mu^2.$$

$$\therefore E(X^2) = \text{Var}(X) + \mu^2 = \sigma^2 + \mu^2 \text{ and}$$

$$E(\bar{X}^2) = \text{Var}(\bar{X}) + \mu^2 = \frac{\sigma^2}{n} + \mu^2.$$

From §4.4, $\text{Var}(\bar{X}) = \frac{\sigma^2}{n}.$